
Part III

Orthopedics of the Lower Extremity

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10.1 Introduction

Analysis of injuries in climbers show that most injuries caused by falls are inflicted onto the feet. These are often standard foot injuries, such as strains and sprains, ankle fractures, and talus or calcaneus fractures. In contrast to injuries onto the upper extremities, about which many studies have been conducted, only few studies exist concerning the lower extremities, especially the feet [1–8]. The tight fit and build of modern climbing shoes can lead to specific injuries and overstrain [1, 9]. Furthermore, the overall number of lower limb injuries in climbers is rising [10, 11]. In

summary of the present literature on the incidence of climbing injuries, it can be stated that injuries to the lower extremities are mostly caused by acute trauma, whereas injuries to the upper extremities are mostly chronic and overstrain injuries [1, 12–14]. These fall-related injuries happen either as a “rock hit trauma” when a climber falls while lead climbing and is pulled against the wall by the rope or as a “ground fall trauma” [1, 13, 15]. Both of these result in either ankle sprains and strains or fractures, mostly of the talus, calcaneus, or the ankle joint [1, 13, 15]. Chronic injuries are mostly due to the tight fit of the climbing shoes as well as their cut, build, and biomechanical design [3, 7, 8, 16]. Bowie et al. [15] as well as others [13, 14, 17, 18] report that roughly 50% of climbing injuries concern the lower extremities. Neuhoof et al. [19] also found an even injury distribution between the upper (42.6%) and lower extremities (41.3%). Killian et al. [8] found a significant correlation between the incidence of ankle sprains and bouldering as well as between ankle sprains and sport climbing. The following chapter will present these acute injuries as well as the chronic, overstrain conditions.

10.2 Modern Climbing Shoes

Until about 40 years ago, most alpine climbing and rock climbing were done in heavy mountain boots. An extra pair of socks was often worn, and the foot was held in a leather cast that protected it but also took away any sensitivity and the need for strength in the toes. Leather boots were replaced by the legendary E.B. in the 1970s, and a new generation of climbing shoes with sticky rubber soles was just around the corner. Since those early days, climbing shoes have become specialized according to the type of rock climbing one wants to do. Attributes such as downturn, the concave shape that places pressure on the toes, and asymmetry, concentrating the pressure on the big toe, are basic elements of modern climbing shoes [8, 9, 20] (Fig. 10.1). These special climbing shoes help the climber to stand on both slabs and edges, using friction



Fig. 10.1 Modern climbing shoe with “downturn” and a “slingshot”

with straight toes and bent toes with precision and proper contact, respectively [7–9, 21, 22]. The majority of foot injuries in climbing result from wearing climbing shoes which are unnaturally shaped or too small in size [7, 8, 21, 22]. The shoe size reduction forces the foot to conform to the shoe and changes the biomechanical position of the foot within the shoe. The foot shortens through supination and contraction of the digits [1, 7, 8, 22]. In front pointing, the proximal and the distal interphalangeal joints are flexed, and the metatarsophalangeal joints are overextended (crimping toes) (Figs. 10.2 and 10.3). Lateral X-rays of feet inside climbing shoes show that the foot weight distribution onto the first and fifth metatarsal head and the heel no longer corresponds to the normal distribution. The foot is front pointing onto the distal toe phalanges [7, 8] (Fig. 10.4). Moreover, the plantar flexion of the metatarsal heads results in a tightening of the plantar fascia [6–8, 22]. High-ability climbers experience more foot deformities and injuries compared to climbers of lower ability due to the common practice of wearing climbing shoes sized smaller than normal streetwear shoes [5, 7, 8]. In 1999, we reported an average shoe size difference between normal shoes and climbing shoes of 2.3 ± 0.73 continental sizes in high-ability rock climbers [mean climbing level 9.7 (UIAA MedCom metric scale)], whereas Killian et al. [8] reported a shoe size difference of one to two continental sizes (mean 1.7) in average climbers [mean climbing level 7.0 (UIAA MedCom metric scale)]. McHenry et al. [5] found ill-fitting and excessively tight footwear in



Fig. 10.2 Radiograph of a foot within a climbing shoe in a standing position



Fig. 10.3 Radiograph of a foot within a climbing shoe in a lateral view. Note the crimping position of the toes and that the forefoot is not resting on the first and fifth metatarsal but on the crimping toes

55 out of 56 rock climbers with a mean difference of climbing shoes to normal shoes of 3.88 UK sizes. Foot pain during activity was also commonplace in 91% of the climbers as well as a mean shoe size reduction of almost four UK

sizes between the climbers' street shoe size and that of their climbing footwear using a calibrated foot/shoe ruler. Furthermore, an unfortunate association between higher climbing specific abilities and the urge for a tighter shoe fit ($p < 0.001$) was reported [5].

For further analysis, acute injuries, mostly from falls and overuse (overstrain) injuries, due to the tight-fitting climbing shoes must be distinguished.

10.3 Chronic Foot Conditions From Climbing Shoes

Modern and "aggressive" (high downturn, very pointy last) climbing shoes are not comfortable, and various studies showed that 80–96% of all climbers report feeling pain when wearing their climbing shoes [1, 5, 7, 8, 22]; simultaneously, studies show that these climbers are bearing the discomfort and pain to achieve a better climbing ability [1, 5, 7–9, 22].

Kilian et al. [8] surveyed 100 climbers through a questionnaire on foot conditions. Seventy-three were male, 27 were female, and the mean climbing level was 5.10 US (7.0 UIAA [26]). Eighty-one percent of the climber's complained of discomfort from the climbing shoes. Pain in the toes or hallux valgus was the most frequent complaint. The authors found a significant correlation between the climbing ability and extent of shoe size reduction; however, they found no correlation between the extent of shoe size reduction or skill level and the incidence of pain or discomfort. Sixty-five percent of the surveyed climbers experienced either tingling or numbness, a condition also frequently mentioned by Peters [22] and Largiadèr and Oelz [25]. These neurological complaints and neuromas are known to be greatly exacerbated through constricting shoes [27]. Van der Putten et al. [7] also frequently found mycosis, splinter hemorrhage, nail fractures, hallux valgus, bunions, and callosities. Nevertheless, as the focus of this publication was on the biomechanical development of a climbing shoe, exact numbers of the distribution of these conditions were not given [1].

Fig. 10.4 Radiograph of feet with and without climbing shoe in a standing position. Note how the climbing shoes presses the large toe into a hallux valgus position



The shoe size reduction forces the foot to conform to the shoe and changes the biomechanical position of the foot within the shoe. The foot shortens through supination and contraction of the digits [7, 8, 22]. In front pointing, the proximal and especially the distal interphalangeal joints are flexed, and the metatarsophalangeal joints are overextended (crimping toes). This causes a deviation of the distal end of the metatarsals [9]. Repetition of the biomechanical non-physiologic position can lead to a hallux valgus deformity [1, 5, 7, 22]. Also, the plantar fascia is under excessive tension [1, 5, 7, 22]. Schöffl et al., Killian et al., and McHenry et al. used radiographic analysis of feet within climbing shoes to show that these kind of shoes press the foot into a hallux valgus-like position [1, 5, 8], with an increase of the hallux valgus angle of about 7° [1] (Fig. 10.4). Fifty-three percent of sport climbers showed a unilateral hallux valgus deformity, and 20% showed a bilateral hallux

valgus deformity in climbers who had been climbing for more than 12 years at a mean climbing level of UIAA metric 9–66 [4]. This rate of a hallux valgus is much higher than in an age-matched comparison group of non-climbers, in which a prevalence of 4.5% could be found [1]. Killian et al. [8] found an average incidence of a hallux valgus of 34% in climbers of all difficulty levels but in 53% of the group of climbers in the 5.12 range (UIAA metric 8.3–9.0).

As the climbing difficulty level increases, the demand that climbing shoes fit tightly and put extreme pressure onto the toes rises because the footholds become smaller [1]. Thus, high-level climbers are more prone to overstrain injuries of the feet [1, 5, 7–9]. In our own study, we evaluated 30 male high-level climbers with a mean climbing training time of 12.3 h per week and a mean climbing level of UIAA metric 9.66 [4]. The respective shoe seize difference was 2.3 sizes (European size). All climbers showed callosities

and pressure marks on the dorsal aspect of the toes. The incidence was the highest onto the large toe (100%) and gradually decreased from toe to



Fig. 10.5 Typical callosity in a 15-year-old female climber

toe in the direction of the small toe (8%) [4] (Fig. 10.5).

Table 10.1 shows the distribution of chronic overstrain pathologies in this study.

A correlation of chronic deformities of the feet and climbing training intensity and climbing level is most likely, especially if the results of our study [4] are compared to that of Largiadèr and Oelz [25]. In the latter study, the authors only found 28% chronic deformities of the toes and 21% of the feet in climbers who had significantly fewer training hours and a distinctly lower climbing level than the athletes from our study [1].

Most of the previously mentioned chronic pathologies can be diagnosed through a thorough clinical examination and radiography. An MRI is only necessary in rare cases.

Depending on the extent of complaints and the deformity, surgical therapy may be necessary (corrective osteotomy of the first metatarsal) in addition to conservative treatment (e.g., adapted footwear, hallux night storage track) [23]. Moreover, in recent years, we have seen an increasing number of climbers with a hallux rigidus condition, even cases requiring a surgical therapy (cheilectomy or fusion) [20]. Rock climbing is still possible after this procedure; nevertheless, the risk of osteoarthritis in the adjunct joints increases [1] (Fig. 10.6). Furthermore, preexisting enchondroma or bone

Table 10.1 Foot deformities in high level climbers (n = 30) [4]

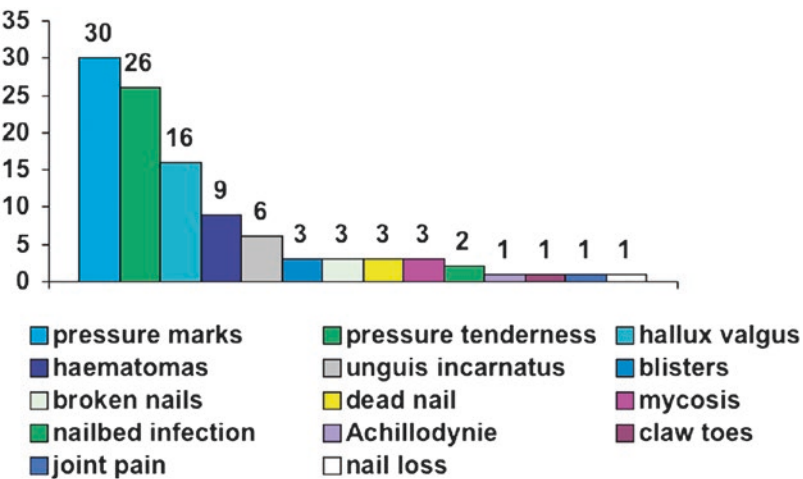




Fig. 10.6 Osteoarthritis auf the large toes base and middle joint in a 35-year-old climber

cysts can lead to increased pain in a tight climbing shoe [1, 5, 7, 8, 18] (Fig. 10.7).

In summary, most climbers are willing to endure some pain in their climbing shoes and accept possible long-term consequences. Climbing ambition and the fun in doing hard moves seem to be too important to the athletes [1, 9] (Fig. 10.8).

Chronic Feet Deformities in Climbers

- **History:** Tight-fitting climbing shoes with toes in a crimping position
- **Onset:** Chronic
- **Clinical:** Pain, callosity, bunions, subungual hematoma, hallux valgus, hallux rigidus
- **Diagnostic:** Clinical finding, radiography
- **Therapy:** Mostly conservative
- **Outcome:** Depending on the shoe size used



Fig. 10.7 Symptomatic enchondroma of the large toe in a female climber



Fig. 10.8 Comparison of a foot with and without climbing shoe

10.4 Traumatic Feet Injuries

In addition to the chronic overstrain injuries caused by tight-fitting climbing shoes, acute injuries, mostly from falls, also need to be evaluated.

Falls are the most common cause for injuries to the lower extremities in rock climbing [15, 19, 28–30]. Thereby, a distinction needs to be made between two different types of falls: a wall-collision fall and a ground fall [20]. In a wall-collision fall, the climber hits the wall in an approximately horizontal plane, whereas during a ground fall, the climber experiences the impact in a mainly vertical plane [20].

In a wall-collision fall, the climber takes a fall while lead climbing. When falling, the tensioning of the rope pulls the climber either to or against the rock face [1, 5, 8, 18, 31]. The force of the impact against the rock face varies in accordance with fall height, belay technique, and rope stretch [20] (Fig. 10.9). Contrary to how it may appear, short falls with a high impact factor (generally not far from the ground or the belay) can be the most dangerous. In those cases, the climber is actually pulled back onto the rock, and the angle at which he or she



Fig. 10.9 A long fall with slack rope, reducing the impact



Fig. 10.10 Ground fall trauma

extends the foot (or arm) upon impact creates a lot of stress to this particular location. Additionally, assisted-braking belay devices such as the Gri-Gri® (Petzl, France) and Sirius® (TRE, Germany) can increase the fall factor [20]. Typical injuries are contusions or compound fractures [20].

The other pathomechanism, the so-called ground fall, is even more common, especially in bouldering (Fig. 10.10). In a ground fall, the impact of the foot onto the ground causes the trauma. A ground fall from a height of as little as 3 meters can already cause serious fractures of the calcaneus, talus, and the ankle joint (Fig. 10.11) [20]. Other ground fall injuries include sprains and ligamentous injuries [13, 14, 17, 20, 32, 33]. These injuries are increasing with the growing popularity of bouldering as a sub-discipline in climbing [10, 34], but they are also fairly common with sport climbers who fall before reaching the first bolt or who are dropped by their belayer [20]. A typical injury mechanism in indoor climbing, especially bouldering, has been a fall onto the mats with the foot stuck in the intermediate section between two mats [28, 29, 35–38]. This has happened quite often, partly because the first indoor



Fig. 10.11 Talus fracture after a climbing fall



Fig. 10.12 Modern indoor bouldering gym with perfect mat covering and no intersections between mats. (Photo: Enrico Haase)

bouldering walls in the late 1980s were small places with old mattresses as protection. Since then, this issue has been addressed by the Medical Commission of the UIAA [38] (International Mountaineering Association), and the International Federation of Sport Climbing (IFSC) has implemented a rule that these intersections must be closed in competitions [35]. Furthermore, this also has become a standard in modern bouldering gyms (Fig. 10.12). In addition to mats without gaps, indoor bouldering gyms implement further safety measures such as innovative wall design to decrease the injury rate. Despite being quite rare, falls onto other boulderers must also be considered regarding injury prevention [9]. For example, in some gyms,



Fig. 10.13 “Crashpad belay” in outdoor bouldering. (Photo: Enrico Haase)

kids run around without parental supervision, during which severe injuries can happen. While kids play, they do not care if someone is climbing above them and could fall onto them. Close supervision and the education of and by the parents are essential. An indoor climbing wall should not be seen as a children’s playground, especially since most gyms provide designated children’s areas [1, 5, 8, 9, 12, 13, 16, 18, 31, 39].

Another risk while bouldering results from items such as chalk bags or water bottles lying underneath the boulders; for example, one climber tore their cruciate ligament at a World Cup after falling onto their own chalk bag.

For bouldering outdoors, which has become very popular in an increasing community, climbers use “crash pads” (portable multiple foam layer mats) as protection and “spotting” to position falling climbers onto these mats. Falls next to or in between the mats also happen outdoors (Fig. 10.13). Most injuries in outdoor bouldering are sprains and strains onto the ankles, as well as talus, calcaneus, or ankle fractures [9, 12, 16, 32, 33, 40, 41]. In rope climbing, ground falls are comparably rare and usually caused by a fall before clipping the first bolt or belay failure [1, 9, 12, 16, 18, 31].

Many studies have already investigated the injury incidence to the lower leg and have reported inhomogeneous injury incidences. Neuhofer et al. [19] evaluated 1962 sport climbers and their injuries over a 5-year period and found a total of 699 injuries. Out of these, 345 (41.3%) were to a lower extremity; 29.2% were on the feet. Most of the fractures (48.6%) and most of all contusions

(42.5%) were to the feet, while most ligament injuries and tendon injuries were to the fingers. Ankle sprains are frequent in boulderers and sport climbers with a significant correlation [8] to these climbing activities. Killian et al. [8] reported an incidence of 0.23 for ankle sprains and 0.03 for foot fractures in rock climbers. Nelsen et al. [13] found that ankle injuries accounted for 19.2% of all climbing injuries; Buzzacott et al. [14] found in a consecutive study that the number increased to 27%. Backe et al. [17] even reported that out of all acute climbing injuries, 50% involved the foot, toe, and ankle. The most common type was a ligament injury, followed by contusions and lacerations. Gerdes et al. [42] also reported that sprains and strains were most common in the fingers (31.9%) and the ankle (23%). Fractures were most common to the foot (24.6%) and the ankle (22.1%). Especially in older athletes, even short falls can lead to foot and ankle fractures caused by the increased brittleness of the bone that occurs with aging [43, 44]. In these cases, even minor falls may result in a serious injury [1, 12, 16, 18, 40] [9, 45].

Typical traumatic feet injuries in rock climbers are [13, 19, 20, 28, 36, 40, 42, 46]: (1) contusion, (2) calcaneus fracture, (3) talus fracture, (4) ankle fracture, and (5) ankle sprain with lateral ligament injury. All of these injuries show an acute onset of symptoms (e.g., pain, swelling, impairment of mobility), and the diagnosis is made by a thorough clinical examination and radiography. In case of an unclear finding, a CT or MRI can be necessary to detect the complete extent of the injury.

The resulting typical injuries should undergo treatment according to standard orthopedic trauma guidelines. In case of displaced fractures, this usually means operative reconstruction (open reduction and internal fixation).

In summary, the literature reports that most injuries in rock climbing are either chronic or occur because of a hard move, leading to pathologies of the upper extremities. Nevertheless, the most serious fractures and sprains are to the feet and ankle joint [1, 10, 14, 16, 33, 39, 40, 42]. As previously discussed, the therapy follows the standard recommendations of the trauma and orthopedic surgical societies.

In analyzing these injuries, one other concomitant effect must be considered. The tight-fitting and non-cushioned climbing shoes described earlier increase the impact and severity of falls because of the change of the foot's biomechanical position within the shoe. The feet are not resting on the first and the fifth metatarsal head but on the toes in a crimp position [4] (Fig. 10.3). This leads to a different, nonphysiologic impact in a fall. Thus, strains and sprains happen more frequently in comparison to falls when normal shoes or mountain boots are worn [1, 5, 7–9, 17, 18].

Traumatic Foot Injuries

- **History:** Ground fall or rock-collision trauma
- **Onset:** Acute
- **Clinical:** Pain, swelling, hematoma, deformity
- **Diagnostic:** Clinical finding, radiography, optional CT/MRI
- **Therapy:** In accordance with orthopedic trauma guidelines
- **Outcome:** Depending on the diagnosis

10.5 Peroneal Tendon Dislocations

Acute peroneal tendon dislocations are rare medical conditions. Although this condition is most often seen in skiers [47–50], rock climbers can also be affected by peroneal tendon and tibialis posterior tendon dislocations [1]. Evaluating 908 climbing injuries over 4 years (2009–2012), we diagnosed three climbers with this pathology [2] (Figs. 10.14 and 10.15). All patients reported a snap of the peroneal retinaculum while having the foot in a plantar flexion with maximum inversion and applying high tension onto the large toe while climbing. These acute peroneal tendon dislocations can be interpreted similarly to pulley injuries in the fingers [51]. With a high-strength impact onto the superior peroneal retinaculum, which serves an analogous function as the pulley system of the fingers, the retinaculum, also enhanced by the calcaneofibular ligament, fails, and the peroneal tendons dislocate.



Fig. 10.14 High stress onto the peroneal tendons standing on tiptoe with inward rotation of the ankle



Fig. 10.16 Surgical reconstruction of the peroneal retinaculum with a periosteal flap



Fig. 10.15 Peroneal tendon dislocation in a climber

In all three cases, the peroneal tendon groove of the fibula was of normal shape; therefore, there was no anatomical predisposition.

Clinically, the patients complain of an acute onset of symptoms. In addition to pain and swelling on the lateral aspect of the upper ankle joint and an impairment of mobility, the peroneal tendon might be visible and palpably dislocated over the lateral malleolus.

The diagnosis is made by a thorough clinical examination and ultrasound, which allows a dynamic imaging of the dislocated tendon. In case of an unclear finding, an MRI can be necessary to detect the complete pathology and determine the shape of the peroneal groove.

In general, literature recommends a nonsurgical approach if the dislocation is treated immediately and the tendons stay in their normal position within a cast [47, 48].

Nevertheless, in active patients, especially in “foot-dependent” sports, surgical repair is recommended [47, 48]. Surgical techniques are mostly described for chronic subluxation and include a peroneal groove deepening, a tenoplasty, or a bone block transfer [47–50, 52]. A direct repair of the ruptured structures is only possible in acute cases, sometimes in combination with a periosteal flap [47, 52]. We performed a surgical repair in all of the previously mentioned cases (Fig. 10.16). In two cases, a direct repair was possible, and in one case, an additional periosteal flap was necessary. After surgery, the area was immobilized for 6 weeks with a cast. Climbing activity was allowed 8 weeks after injury. All athletes regained their full climbing ability with no reoccurrence or consisting problems. Overall, we recommend a surgical repair of dislocations of peroneal as well as tibialis posterior tendons in climbers.

Peroneal Tendons Dislocation

- **History:** High stress onto the foot in inward rotation
- **Onset:** Acute

- **Clinical:** Pain, visible and palpable dislocated tendon
- **Diagnostic:** Clinical finding, ultrasound, MRI optional (mostly to determine the shape of the peroneal groove)
- **Therapy:** Surgical
- **Outcome:** Good

10.6 Conclusion

Considering the high incidence of feet pathologies in rock climbers, it is astonishing that so far, very little scientific research has been carried out on this topic. With numbers of up to 90% of climbers with chronic foot conditions, more research needs to be done in this respect (1). To reduce these complaints, climbers need to be advised not to wear their shoes too tight and to have additional, loose-fitting training shoes [20]. The climbing shoe industry should also be involved. New fitting strategies with less stretchy outer materials and biomechanically adjusted constructions are promising approaches in this area [1]. New shoes should have an inner lining to reduce bunions and callosities while still guaranteeing a good perception of the rock. While high-level climbing athletes may require a tight-fitting shoe, the vast majority of recreational climbers need the opposite: a comfortably fitting shoe which supports the foot and does not hurt [1].

References

- Schöffl V, Küpper T. Feet injuries in rock climbers. *World J Orthop.* 2013;4(4):218–28.
- Heid A, Popp D, Schöffl V. Traumatic peroneal tendon dislocations in rock climbers “the climbers pulley lesion of the foot”. *Med Sport.* 2013;17(4):188–92.
- Morrison AB, Schöffl VR. Physiological responses to rock climbing in young climbers. *Br J Sports Med.* 2007;41(12):852–61; discussion 61.
- Schöffl V, Winkelmann HP. [Foot deformations in sport climbers] Fußdeformatitäten bei Sportkletterern. *D Z Sportmed.* 1999;50:73–6.
- McHenry RD, Arnold GP, Wang W, Abboud RJ. Footwear in rock climbing: current practice. *Foot (Edinb).* 2015;25(3):152–8.
- Peters P. Orthopedic problems in sport climbing. *Wilderness Environ Med.* 2001;12(2):100–10.
- van der Putten EP, Snijder CJ. Shoe design for prevention of injuries in sport climbing. *Appl Ergon.* 2001;32(4):379–87.
- Killian RB, Nishimoto GS, Page JC. Foot and ankle injuries related to rock climbing. The role of footwear. *J Am Podiatr Med Assoc.* 1998;88(8):365–74.
- Lutter C, Schöffl V. Trendsportart Bouldern—Verletzungen und chronische Schäden der unteren Extremität. In: Faulhaber M, Schobensberger W, Schobensberger B, Sumann G, Domej W, editors. *Jahrbuch 2015—Österreichische Gesellschaft für Alpin- und Höhenmedizin.* Innsbruck: Österreichische Gesellschaft für Alpin- und Höhenmedizin; 2015. p. 43–56.
- Lutter C, Tischer T, Hotfiel T, Frank L, Enz A, Simon M, et al. Current trends in sport climbing injuries after the inclusion into the Olympic program. Analysis of 633 injuries within the years 2017/18. 2020.
- Lutter C, Tischer T, Cooper C, Frank L, Hotfiel T, Lenz R, et al. Mechanisms of acute knee injuries in bouldering and rock climbing athletes. *Am J Sports Med.* 2020;48(3):730–8.
- Schöffl V, Morrison A, Schöffl I, Küpper T. The epidemiology of injury in mountaineering, rock and ice climbing. *Med Sport Sci.* 2012;58:17–43.
- Nelson NG, McKenzie LB. Rock climbing injuries treated in emergency departments in the U.S., 1990–2007. *Am J Prev Med.* 2009;37(3):195–200.
- Buzzacott P, Schöffl I, Chimiak J, Schöffl V. Rock climbing injuries treated in US Emergency Departments, 2008–2016. *Wilderness Environ Med.* 2019;
- Bowie WS, Hunt TK, Allen HA Jr. Rock-climbing injuries in Yosemite National Park. *West J Med.* 1988;149(2):172–7.
- Neuhof A, Hennig FF, Schöffl I, Schöffl V. Injury risk evaluation in sport climbing. *Int J Sports Med.* 2011;32(10):794–800.
- Backe S, Ericson L, Janson S, Timpka T. Rock climbing injury rates and associated risk factors in a general climbing population. *Scand J Med Sci Sports.* 2009;19(6):850–6.
- Schöffl V, Popp D, Küpper T, Schöffl I. Injury trends in rock climbers: evaluation of a case series of 911 injuries between 2009 and 2012. *Wilderness Environ Med.* 2015;26(1):62–7.
- Neuhof A, Hennig FF, I. S, V. S. Injury risk evaluation in sport climbing. *Int J Sports Med.* 2011;32(10)
- Hochholzer T, Schöffl V. One move too many. 2nd ed. *Ebenhausen: Lochner Verlag;* 2006.
- Schöffl V, Winkelmann HP. Foot deformations in Sport climbers. *Ö J Sportmed.* 1999;3:58.
- Peters P. Nerve compression syndromes in sport climbers. *Int J Sports Med.* 2001;22(8):611–7.
- Santrock RD, Smith B. Hallux valgus deformity and treatment: a three-dimensional approach: modified technique for lapidus procedure. *Foot Ankle Clin.* 2018;23(2):281–95.

24. Schöffl V, Morrison A, Hefti U, Ullrich S, Küpper T. The UIAA Medical Commission injury classification for mountaineering and climbing sports. *Wilderness Environ Med.* 2011;22(1):46–51.
25. Largiadier U, Oelz O. An analysis of overstrain injuries in rock climbing. *Schweiz Z Sportmed.* 1993;41(3):107–14.
26. Schöffl V, Morrison AB, Hefti U, Schwarz U, Küpper T. The UIAA Medical Commission Injury Classification for Mountaineering and Climbing Sports. *Wilderness Environ Med.* 2011;22(1):46–51.
27. Bouche RT, Sullivan K, Ichikawa DJ. Athletic injuries. In: Oloff LM, editor. *Musculoskeletal disorders of the lower extremities*. 1st. 1st ed. Philadelphia: WB Saunders; 1994. p. 34.
28. Schöffl V, Morrison A, Schöffl I, Küpper T. Epidemiology of injury in mountaineering, rock and ice climbing. In: Caine D, Heggie T, editors. *Medicine and sport science—epidemiology of injury in adventure and extreme sports*. Vol. 58: Karger; 2012. p. 17–43.
29. Schöffl V, Morrison AB, Schwarz U, Schöffl I, Küpper T. Evaluation of injury and fatality risk in rock and ice climbing. *Sport Med.* 2010;40(8):657–79.
30. Mosimann U. Notfälle beim Eisklettern. *Ther Ber.* 2006;4:70–3.
31. Schöffl VR, Hoffmann G, Küpper T. Acute injury risk and severity in indoor climbing—a prospective analysis of 515,337 indoor climbing wall visits in 5 years. *Wilderness Environ Med.* 2013;24(3):187–94.
32. Jones G, Schöffl V, Johnson MI. Incidence, diagnosis, and management of injury in sport climbing and bouldering: a critical review. *Curr Sports Med Rep.* 2018;17(11):396–401.
33. Jones G, Asghar A, Llewellyn DJ. The epidemiology of rock-climbing injuries. *Br J Sports Med.* 2008;42(9):773–8.
34. Lutter C, Tischer T, Schöffl V. Olympic competition climbing: the beginning of a new era. *British Journal Sportsmed* (in press). 2020;
35. Schöffl V, Schöffl I. Competition climbing. *Sport ortho Sport trauma.* 2012;28:22–8.
36. Schöffl V, Küpper T. Injuries at the 2005 World Championships in Rock Climbing. *Wilderness Environ Med.* 2006;17:187–90.
37. Schöffl V, Winkelmann HP. [Accident statistics at “indoor climbing walls”] *Unfallstatistik an “Indoor-Kletteranlagen”*. Sportverletz Sportschaden. 1999;13(1):14–6.
38. Schöffl V, Schlegel C. Statement Competition Climbing: www.uiiaa.ch; 2000. www.uiiaa.ch.
39. Schöffl VR, Küpper T. Injuries at the 2005 World Championships in Rock Climbing. *Wilderness Environ Med.* 2006;17(3):187–90.
40. Josephsen G, Shinneman S, Tamayo-Sarver J, Josephsen K, Boulware D, Hunt M, et al. Injuries in bouldering: a prospective study. *Wilderness Environ Med.* 2007;18(4):271–80.
41. Jones G, Johnson M. A critical review of the incidence and risk factors for finger injuries in rock climbing. *Curr Sports Med Rep.* 2016;15(6):400–9.
42. Gerdes EM, Hafner JW, Aldag JC. Injury patterns and safety practices of rock climbers. *J Trauma.* 2006;61(6):1517–25.
43. Hartsock LA, Feagin JA Jr, Ogilvie BC. Climbing and the older athlete. *Clin Sports Med.* 1991;10(2):257–67.
44. Lutter C, Hotfiel T, Tischer T, Lenz R, Schöffl V. Evaluation of rock climbing related injuries in older athletes. *Wilderness Environ Med.* 2019;30(4):362–8.
45. Schöffl V, Lutter C. The “Newbie” syndrome. *Wilderness Environ Med.* 2017;28(4):377–80.
46. Jones G, Asghar A, Llewellyn DJ. The epidemiology of rock climbing injuries. *Br J Sports Med.* 2007;42:773–8.
47. Arrowsmith SR, Fleming LL, Allman FL. Traumatic dislocations of the peroneal tendons. *Am J Sports Med.* 1983;11(3):142–6.
48. Brage ME, Hansen ST Jr. Traumatic subluxation/dislocation of the peroneal tendons. *Foot Ankle.* 1992;13(7):423–31.
49. Mendicino RW, Orsini RC, Whitman SE, Catanzariti AR. Fibular groove deepening for recurrent peroneal subluxation. *J Foot Ankle Surg.* 2001;40(4):252–63.
50. Kollias SL, Ferkel RD. Fibular grooving for recurrent peroneal tendon subluxation. *Am J Sports Med.* 1997;25(3):329–35.
51. Schöffl V, Hochholzer T, Winkelmann HP, Strecker W. Pulley injuries in rock climbers. *Wilderness Environ Med.* 2003;14(2):94–100.
52. Tan V, Lin SS, Okereke E. Superior peroneal retinaculoplasty: a surgical technique for peroneal subluxation. *Clin Orthop Relat Res.* 2003;410:320–5.